

Martin Nunction

Martin Nunction(a pun for Notation/Function), Is a large number notation created by Martin Wanralfs or UpBeatSquirrel925 in Fandom wiki.

A.Definitions

1.Martin Bunction (MB)

Martin Bunction is the very first function in Martin nunction, written as:

$$B(a,b,c)$$

Let a be any non-zero finite positive integer, a is the argument, a is the first entry.

Let b be any non-zero finite positive integer, b is the amount of (c-th) hyperoperation, b is the second entry.

Let c be any non-zero finite positive integer, c is the hyperoperation level, c is the third entry.

General Martin Bunction expansion:

$$B(a,b,c) = a \uparrow^b a = a \uparrow^{b-1} (a \uparrow^{b-1} a)$$

Right associatively, using a value as the amount of iteration.

(Note: octothorpe symbol means unchanged remainder of an array and hyperoperation.)

$$B(a,b,c) = a \uparrow^c a$$

General Martin Bunction leveling down:

$$B(a,1,c) = a \uparrow^c a = a \uparrow^{c-1} a, \text{ or}$$

$$A \uparrow^c n = a \uparrow^{c-1} n$$

Where b = 1, b value replaced by right operand a, while c is subtracted by 1, moving to (c-1)-th hyperoperation.

Examples:

$$B(1,b,c) = 1 \uparrow^b 1 = 1$$

$$B(1,1,1) = 1 \uparrow^1 1 = 1^1 = 1$$

$$B(1,n,n) = 1 \uparrow^n 1 = 1$$

Where c = 1 is the Knuth's up arrow hyperoperation, a = 1 is the argument and hyperoperation expansion operand, and b = 1 is the amount of current level hyperoperation, a becomes the result ignoring the second and third entry iff a = 1.

$$B(2,2,2) = 2 (2)^2 2 = 2 (2)^{12} = 2 \uparrow^2 2 = 2 \uparrow 2 = 2^2 = 4.$$

$$B(2,b,c) = 2 (c)^b 2 = 2 (c)^{b-1} 2 = 2 (c)^{b-2} 2 \dots$$

Operation will terminate to 4 such that $a = 2$.

$$B(3,2,1) = 3 \uparrow^2 3 = 3 \uparrow (3 \uparrow 3) = 3 \uparrow 27 = 7,625,597,484,987.$$

2.Martin Array Bunction (MAB)

The next level in Martin Nunction, extending the array length of Martin Bunction, written as

$$B(a,b,c,d,e,\#)$$

(Note: octothorpe symbol means unchanged remainder of an array and hyperoperation.)

Where array entry must be a non-zero finite positive integer or an array.

General 4 Entry Expansion:

$$B(a,b,c,d) = B(a,b,(a,b-1,c,d),d-1)$$

General Pairing rule for Entry >3 where there's a number 1 followed by a number 2, and where 1 followed by a number > 2, scanning right to left:

$$B(a,1,2,\#) = B(a,3,1,\#)$$

$$B(a,1,c,\#) = B(a,(a,(\dots\#\dots),c-1,\#),c-1,\#)$$

Where b is now replaced with an entire array nested into b entry c times, subtracting c by 1 on the outer array and all inner arrays

General n entries:

$B(a,b,c,d,\#) \rightarrow B(a,b,c,1,\#)$, Entries after d are for applying pairing rule such that any entry except for a = 1.

Example:

$$B(3,2,2,2) = B(3,2,(3,1,2,2),1) \rightarrow B(3,2,(3,1,2,2)), \text{ remove trailing 1, according to rule 3.}$$

→ Applying pairing rule, for pair $b=1$ and $c=2$, (3,1).

$$B(3,2,(3,3,1,2))$$

→ Applying pairing rule, for pair $c=1$ and $d=2$, (3,1).

$$B(3,2,(3,3,3,1)) \rightarrow B(3,2,(3,3,3)), \text{ remove trailing 1.}$$

→ Let $X = B(3,3,3)$

$$\rightarrow B(3,2,X) = 3 (X)^2 3.$$

3.Martin Array Nunction (MAN)

The next level in Martin Nunction, an array of arrays, where:

$$\backslash a \backslash = B(a,a,a,\dots(B(a,a,a,a))\text{entries}\dots\#)$$

Martin Bunction array filled with a, and with B(a,a,a,a) amount of entries

$$\backslash a,b \backslash = B(a,a,\dots(B(a,a,\dots(B(b,b,b,b))\text{entries}\dots))\text{entries}\dots\#)$$

Martin Bunction array filled with a, and with Martin Bunction array filled with a, which have B(b,b,b,b) amount of entries.

General n entries in MAN:

$$\backslash a,b,c,\#,\backslash k \backslash = B(a,a,\dots(B(a,a,\dots(B(b,b,\dots((\dots(B(k,k,k,k))\text{entries}\dots))\text{entries}\dots))\text{entries}\dots\#)$$

Martin Bunction array filled with a, and with Martin Bunction array filled with a amount of entries, with each array entry iteration have it's own entry iterations.

Where entries must be non-zero finite positive integer for MAN.

4.Bangs (!)

An Array operator of MAN, the first priority Operator, written in the operator slots which are second to last entry in MAN.

General entry ≤ 3 with single or more bangs:

General Linear MAN array with single bang:

$$\backslash a,! ,b \backslash = \backslash a,a,\dots\# \backslash$$

MAN linear array with $\backslash b \backslash$ Entries.

$$\backslash a,b,! ,c \backslash = \backslash a,a,\dots\# \backslash$$

MAN linear array with $\backslash b,c \backslash$ entries.

General linear MAN array with n bangs:

$$\backslash a,!^{(n)},b \backslash = \backslash a,a,\dots\#,\dots,!^{(n-1)},k \backslash$$

MAN linear array with (n-1) bangs and $\backslash b \backslash$ entries.

$$\backslash a,b,!^{(n)},c \backslash = a,a,\dots\#,\dots,!^{(n-1)},k \backslash$$

MAN linear array with (n-1) bangs and $\backslash b,c \backslash$ entries.

>3 entry with bangs recursion:

$$\backslash a,b,c,! ,d \backslash = \text{Let } X = \backslash a,b,! ,c \backslash, \text{ calculate 3 Entry with single bang first.}$$

$$\rightarrow \backslash X,X,X,! ,d-1 \backslash.$$

Example :

$\backslash 2,2,2,! ,2\backslash = \text{Let } X = \backslash 2,2,! ,2\backslash$.

$\rightarrow \backslash 2,2,! ,2\backslash = \backslash 2,2,2,2,2\backslash$, calculate $\backslash b,c\backslash$,

$B(b,b,\dots(B(2,2,2,2))\text{entries..}\#)$ MAB with $\backslash c\backslash$ entries = $B(2,2,2,2) = 4$, MAB with 4 entries filled with b.

$\rightarrow B(2,2,2,2) = 4$.

$\rightarrow \backslash 2,2,! ,2\backslash = 4$, replace all entry before the bang with X.

$\rightarrow \backslash 4,4,4,! ,1\backslash$, remove trailing 1, according to rule 3, for bang, the bang is moved to before the new rightmost entry .

$\rightarrow \backslash 4,4,! ,4\backslash$.

$\backslash a,b,c,d,! ,e\backslash = \text{Let } X = \backslash a,b,c,! ,d\backslash$, calculate 4 entry with single bang first.

$\rightarrow \backslash X,X,X,X,! ,e-1\backslash$.

General n entries with single or more bangs:

Let $L(n,!^n) = \text{Entry with Length } n \text{ and } n \text{ bangs}$

For single bang and n Entry:

$\backslash a,b,c,\#,! ,k\backslash = \text{Let } X = \backslash a,b,c,\#,! ,y\backslash$, calculate $L(n-1,!)$ first.

$\rightarrow$ replace every entry before bang with X

$\rightarrow \backslash X,X,X,\#,! ,k-1\backslash$

For n bangs with n length:

$\backslash a,b,\#,!^n(n),k\backslash = \text{Let } X = \backslash a,b,\#,!^n(n),j\backslash$, calculate $L(n-1,!^n)$ first.

5. Tiers (/n/)

The second priority operator.

$\backslash a,/n/,b\backslash$.

General Tiering:

$/n/ = \text{Tier 1 with value } n$, $//n// = \text{Tier 2 with value } n$, (k-th tier(n) = k-th tier with value n, higher tier = less priority.

General tier 1 with $n = 1$ and $n > 2$:

$\backslash a,/1/,b\backslash = \text{Let } X = \text{Result of } \backslash a,b\backslash$.

$\rightarrow \backslash a,!^n(\backslash a,!^n(\backslash a,!^n(\dots\#,\dots),b\backslash),b\backslash),b\backslash$.

Evaluate X, then subtract tier 1 by 1, which becomes X amounts of Bangs, with entire array nested into it's exponent, Where innermost array = $\backslash a, !^X, b \backslash$

$\backslash a, /n, b \backslash = \text{Let } X = \text{Result of } \backslash a, b \backslash,$

$\rightarrow \backslash a, /n-1, !^{\backslash a, /n-1, !^{\backslash a, /n-1, !^{\dots \# \dots}, b \backslash}, b \backslash}, b \backslash$

Evaluate X, then subtract tier 1 by 1, which becomes X amounts of Bangs added to the second rightmost entry as higher priority operator, with entire array nested into it's exponent, Where the innermost array is, $\backslash a, /n-1, !^X, b \backslash$

General Tier 2 and Tier k with value n:

$A, //n, b \backslash = \text{Let } X = \text{Result of } \backslash a, b \backslash.$

$\rightarrow \backslash a, //n-1, /(\backslash a, //n-1, /(\dots \# \dots), b \backslash), b \backslash$

Where innermost array = $\backslash a, //n-1, /X, b \backslash$

Nest entire array X times inside Tier 1 bracket and subtract Tier 2 value by 1, then add lower tier operator on the right side of the higher tier in the operator slots

$\backslash a, (k\text{-th tier}(n)), b \backslash = \text{Let } X = \text{Result of } \backslash a, b \backslash.$

$\rightarrow \backslash a, (k\text{-th tier}(n-1), /(\backslash a, (k\text{-th tier}(n-1)), / \# , b \backslash), b \backslash$

Where innermost array = $\backslash a, (k\text{-th tier}(n-1), (k-1 \text{ tier}(X)), b \backslash.$

Nest entire array into (k-1)-th tier bracket and subtract k-th tier value by 1, then add lower tier operator on the right side of the higher tier in the operator slots.

6. Martin Kunction (MK)

Index of Structure and tiers, written as:

$M_k(m, n)$

Let k = k-th tier structure within Martin Nunction.

Let m = tier within Current tier structure

Let n = amount of current tier within the structure

$M_k(m, n) = \backslash n, (k\text{-th structure tier } m(n)), n \backslash$

Example:

$M_1(1, 3) = \backslash 3, /3, 3 \backslash$

$$M_1(2,2) = \backslash 2, // 2 //, 2 \backslash$$

$$M_2(1,3) = \backslash 3, \{3\}, 3 \backslash$$

$$M_2(2,2) = \backslash 2, \{\{2\}\}, 2 \backslash$$

$$M_3(1,2) = \backslash 2, [2], 2 \backslash$$

General Martin Kunction:

$$M_k(m,n) = \backslash n, (k\text{-th structure tier } m(n)), n \backslash \text{ Let } X = \backslash n, n \backslash,$$

$$\rightarrow \backslash n, (k\text{-th structure tier } m(n-1)), (k\text{-th structure tier } m-1(\backslash n, (k\text{-th structure tier } m(n-1), (k\text{-th structure tier } m-1(\#, n \backslash))), n \backslash$$

Calculate X, subtract k-th structure m-th tier value by 1, then add k-th Structure (m-1)-th tier with value X, nest entire array into it's bracket,

Where Innermost array = $\backslash n, (k\text{-th structure tier } m(n-1), (k\text{-th structure tier } m-1(X), n \backslash$

Leveling Down Structure:

$$\backslash n, (k\text{-th structure tier } 1(1)), n \backslash \text{ Let } X = \text{result of } \backslash n, n \backslash$$

$$\rightarrow \backslash n, (k-1 \text{ structure tier } X(\backslash n, (k-1 \text{ structure tier } X(\#), n \backslash))), n \backslash$$

Calculate X, Level down to (k-1)-th structure with tier X and Value X, nest the entire array into the (k-1)-th structure X-th tier bracket X times, with Innermost array = $\backslash n, (k-1 \text{ structure tier } X(X), n \backslash$

7. Martin K Nunction (MKN)

Self-reference of MK, written as:

$$M^{\{w\}}_k(m,n)$$

Let w = Amount of nesting parameter k.

$$M^{1}_k(m,n) = M_{(M_k(m,n))}(m,n) = \backslash n, ((1) \text{ k-th structure tier } m(n)), n \backslash = \backslash n, ((\backslash n, (k\text{-th structure tier } m(n)), n \backslash) \text{-th structure tier } m(n)), n \backslash$$

$$M^{2}_k(m,n) = M^{1}_{(M^{1}_k(m,n))}(m,n)$$

$$\text{General} = M^w_k(m,n) = M^{(w-1)}_{(M^{(w-1)}_{(M^{(w-1)}_{\#}(m,n)))}(m,n)$$

Subtract w by 1, nest w-1 w times as MK Exponent or w parameter.

B. Rules

Rule 1: array entry and (n) must contain non zero positive integers except for

Operators Slot and Value or amounts.

Rule 2: $\backslash a, b, c, \dots, \text{Operator}, k \backslash$ or $\backslash a, \text{Operator}, b \backslash$ or $\backslash a, [n], \{n\}, n/, !, b \backslash$.

Higher priority operator is written right before last entry, least Priority written after Second last array entry and so on, operator slots are always second to rightmost entry in MAN.

Rule 3: remove trailing 1, $\backslash a, b, c, !, 1 \backslash \rightarrow \backslash a, b, !, c \backslash$ for MAN Bangs, $B(\#, 1) \rightarrow B(\#)$ for MAB.

Rule 4: Prioritize (!) before m-th tier, then k-th structure.

Rule 5: Evaluate Innermost array first.

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